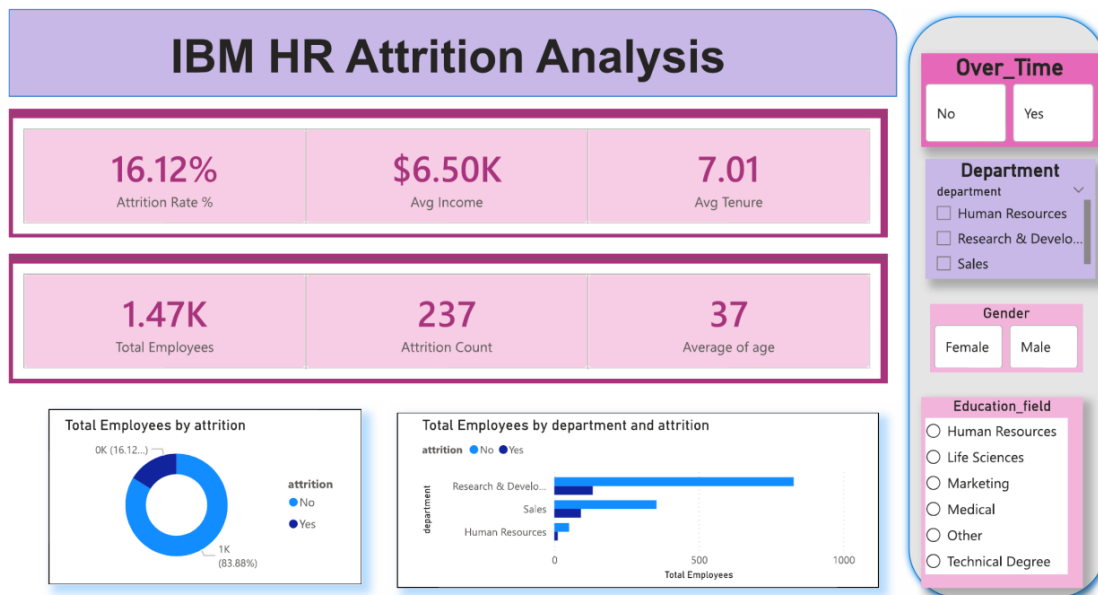


IBM HR Attrition Analysis

Why people leave, and where the company should look first

An end-to-end analytics project · Python · SQL · Power BI



The Power BI dashboard built for this project (page-level view).

Venkata Modem

Data Analytics Portfolio Project

1. The problem I set out to answer

Every time someone resigns, the company pays for it twice. There is the obvious cost of hiring and training a replacement, and then the quieter cost of the knowledge, relationships and momentum that walk out the door with them. For a large employer like IBM, even a small percentage of avoidable departures adds up to a serious bill every year.

So the question driving this project was simple to state and harder to answer: **who is leaving, and is there a pattern behind it that HR can actually act on?** Rather than treat attrition as bad luck, I wanted to see whether the people who leave share recognisable traits, the kind a manager could spot early enough to do something about.

To get there I worked the data through three stages. I cleaned and explored it in **Python**, asked focused business questions of it in **SQL**, and then put the answers in front of a non-technical audience with a **Power BI** dashboard. This report walks through that journey and lands on what I think the findings are really saying.

2. The data I worked with

The analysis uses the well-known **IBM HR Employee Attrition** dataset, a public sample of **1,470 employees** with 35 columns describing each person: their age, role, department, pay, tenure, satisfaction scores, whether they work overtime, and crucially whether they have left the company. It is a fictional dataset created by IBM for analysis practice, so the names and records are not real people, but the structure mirrors what a genuine HR system would hold.

One honest caveat up front: because this is a static snapshot rather than a live feed, everything here describes **association, not proof of cause**. When I say overtime lines up with higher attrition, I mean exactly that. It is a strong signal worth investigating, not a closed case. I have tried to keep that distinction visible throughout.

3. Tools and workflow

Stage	Tool	What happened here
Clean & explore	Python (pandas, seaborn)	Dropped dead columns, checked correlations, profiled who leaves
Question the data	SQL (MySQL)	Nine business questions answered with grouped queries and window functions
Communicate	Power BI	A single interactive page with KPIs, slicers and breakdowns for HR

The cleaned table was loaded into a MySQL database so the SQL layer ran against a proper relational source rather than a flat file, which is closer to how this would work on a real team.

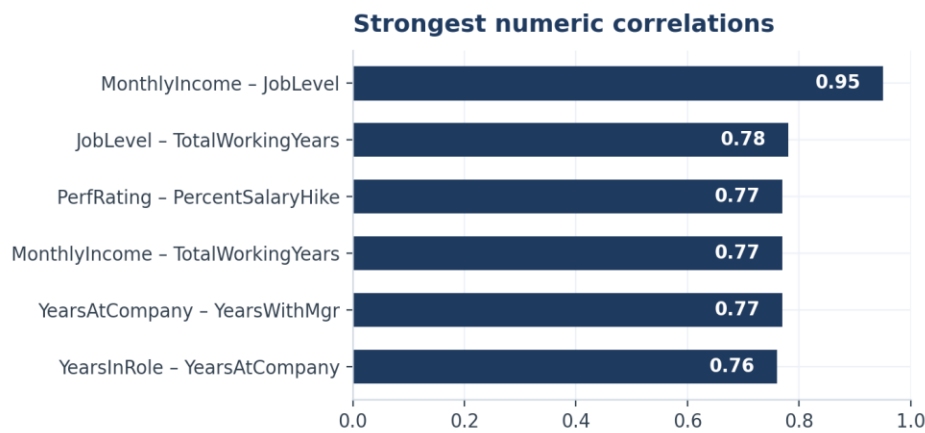
4. Getting the data ready

Good analysis starts with throwing things away. Several columns in the raw file carried no information at all because they held the same value for every single employee, **EmployeeCount**, **Over18** and **StandardHours** among them. Keeping columns like that only adds noise, so they were removed. I also dropped identifier and rate fields (**EmployeeNumber**, **DailyRate**, **HourlyRate**, **MonthlyRate**) that don't help explain behaviour.

After that I standardised every column name to clean lower-case snake_case (so **MonthlyIncome** became **monthly_income**), which makes the later SQL far easier to read and avoids the usual headaches with capital letters and spaces. I also banded **years in current role** into four experience tiers, from Associate up to Principal, so seniority could be compared in plain language instead of raw years.

What the correlations told me early

Before asking any business questions, I looked at how the numeric fields move together. A few relationships stood out immediately, and they mostly confirm common sense, which is a good sign the data is behaving sensibly:



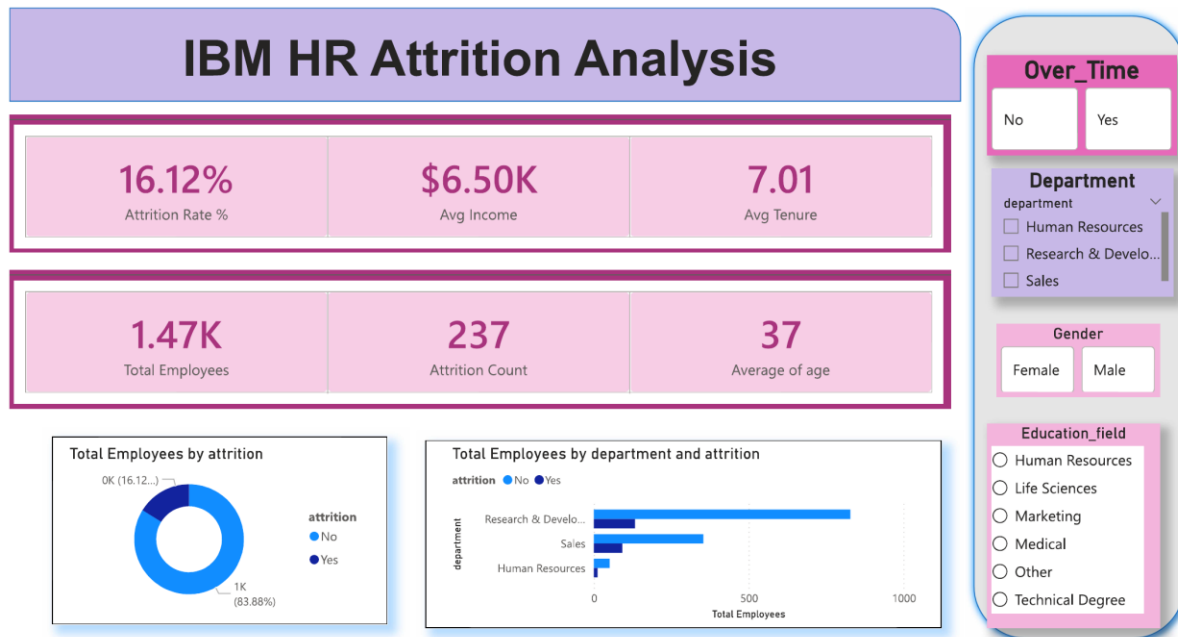
The strongest numeric relationships in the data.

- **Monthly income and job level move almost in lock-step (0.95).** Pay is essentially a function of level, exactly as you'd expect in a banded salary structure.
- **Tenure fields cluster together.** Years at the company, years in role and years with the current manager all rise and fall together, so they partly tell the same story.
- **Performance rating tracks the salary hike (0.77).** Higher-rated people get bigger raises, which is reassuring for the fairness of the reward system.

The practical takeaway: a handful of these fields overlap heavily, so if this ever fed a predictive model, I would not throw all of them in at once. That overlap is something to manage, not ignore.

5. The dashboard at a glance

This is the page I built for HR. The idea was to make the headline numbers readable in five seconds, then let anyone slice them by overtime, department, gender or education field using the filters on the right. No SQL knowledge required.



IBM HR Attrition Analysis — Power BI dashboard.

The numbers that frame everything

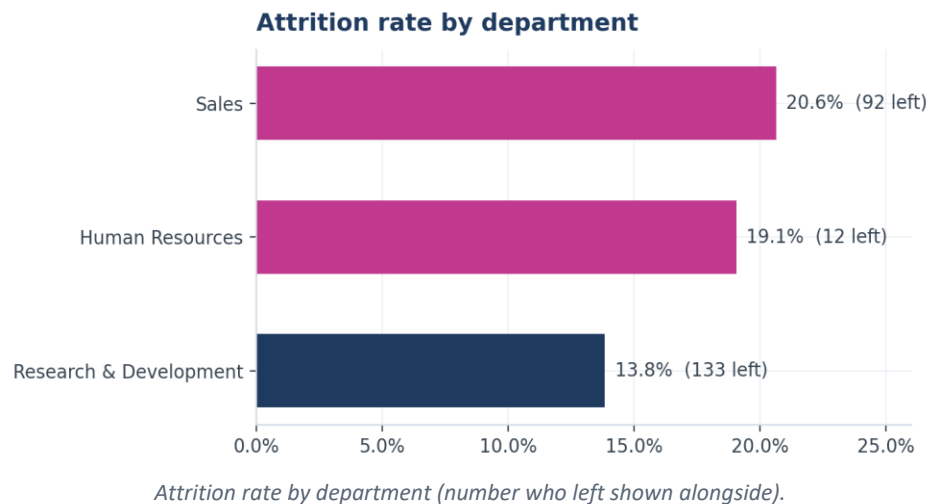
Metric	Value	In plain English
Total employees	1,470	The full workforce in the sample
Attrition count	237	People who have left
Attrition rate	16.12%	Roughly one in six employees
Average monthly income	\$6.5K	Across all staff
Average tenure	7.01 years	Typical time at the company
Average age	37	A fairly experienced workforce

A 16% attrition rate is the anchor for the whole analysis. On its own it sounds moderate. The interesting part is that this average hides pockets where the rate is more than double, and those pockets are where the real story lives.

6. What the analysis actually found

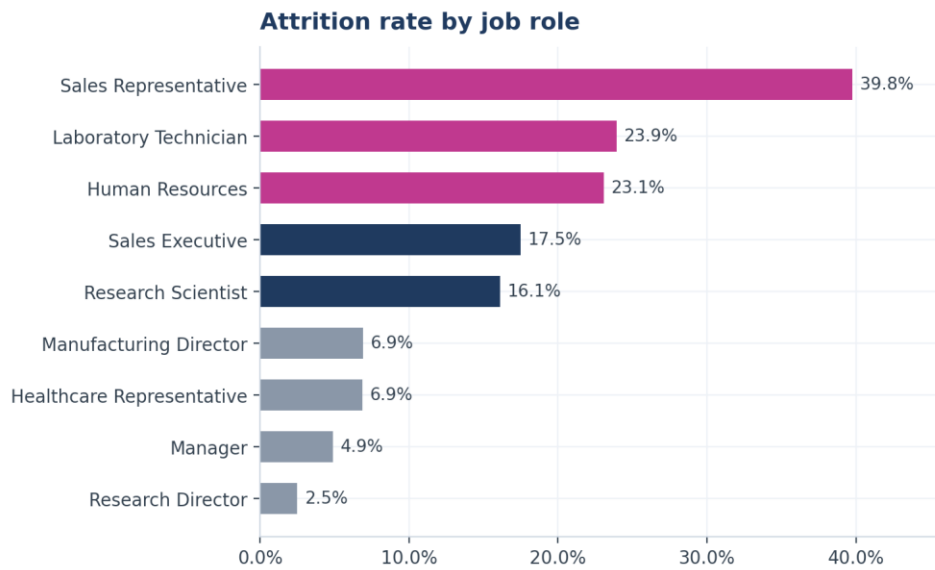
Sales loses people fastest, even though R&D loses more bodies

This is a distinction worth slowing down on. In raw headcount, Research & Development sees the most departures simply because it is by far the biggest department. But **rate** tells a different and more useful story. **Sales has the highest attrition rate at around 21%**, noticeably above the company average, while R&D sits below it at roughly 14%. If you only looked at counts, you would aim your attention at the wrong place.



One role stands out from all the others

Drilling into job roles makes the picture sharper. **Sales Representatives leave at nearly 40%**, almost two and a half times the company average and far ahead of any other role. Laboratory Technicians and Human Resources staff form a second tier in the low-to-mid 20s. At the safe end, Research Directors and Managers barely move, leaving at under 5%.

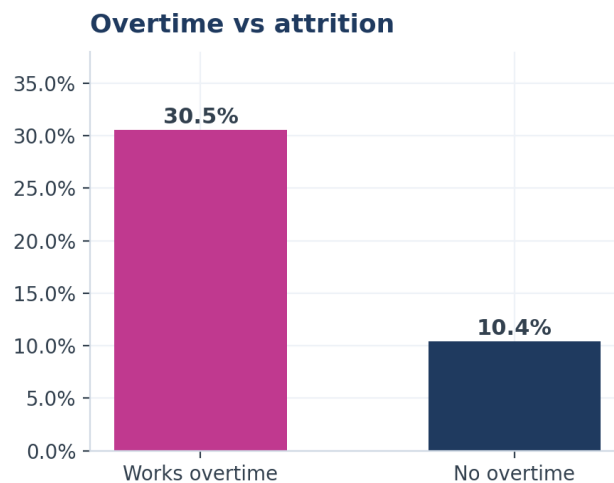


Attrition rate by job role. Pink marks the roles above 20%.

That single bar, Sales Representative, is arguably the most important finding in the whole project. It is a junior, high-pressure, lower-paid role, and the data is shouting that it churns harder than anything else. If HR could only fix one thing, this is where I would point them.

Overtime is the clearest behavioural signal

Of all the factors I looked at, overtime gives the cleanest split. People who regularly work overtime leave at about **30%**, roughly three times the **10%** rate of those who don't. Whatever overtime represents here, whether genuine overload, burnout, or simply the kind of role that demands it, it is strongly tied to people walking away.

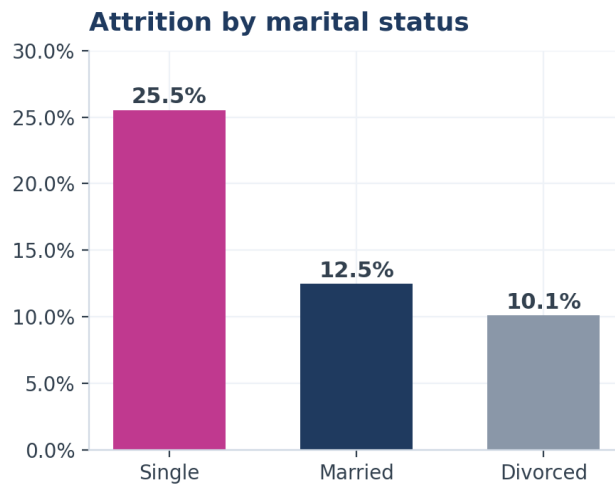


Attrition is about 3x higher among employees who work overtime.

Life stage shows up too

Single employees leave at around **26%**, clearly more than married (12%) or divorced (10%) colleagues. This tends to track with age and life stage rather than marital status being a cause in itself. Younger employees

earlier in their careers are simply more mobile, more open to a better offer, and less anchored by commitments. The pattern is consistent with the role finding above, since junior Sales Representatives skew young.



Attrition by marital status.

Pay matters, but it is the floor that bites

Lower earners leave more often. This is not surprising, but it interacts with everything else: the highest-churn role (Sales Representative) is also one of the lower-paid, and lower-paid people are the most likely to be tempted elsewhere. Pay isn't a standalone villain so much as the thread tying several of these findings together.

7. Putting it together

Step back from the individual charts and a single profile emerges fairly clearly. The employee most at risk of leaving tends to be **younger, single, in a junior or sales-facing role, earning at the lower end, and working overtime**. None of those traits is shocking on its own. What makes the finding useful is that they stack. When several of them land on the same person, the risk compounds, and a Sales Representative working overtime is the textbook example.

That is genuinely actionable, because it tells HR where to spend limited attention. You cannot run a retention programme for everybody. You can run one for the few hundred people who sit at that intersection.

What I'd suggest doing about it

Reading the findings as a to-do list rather than a wall chart, three things stand out:

- **Start with Sales Representatives.** A 40% loss rate in one role is not a workforce-wide problem, it is a focused one. Exit interviews and a look at pay, targets and workload for that specific group would likely pay for itself.
- **Treat overtime as an early-warning flag.** It is one of the few fields that splits leavers from stayers so cleanly. Tracking who is consistently over hours gives managers a list of people to check in on before they reach the door.
- **Watch the lower pay bands.** The combination of low income and a junior role is where retention spending goes furthest, because that is where people are both most likely to leave and most movable by a modest intervention.

Where this stops short (and what I'd do next)

I want to be straight about the limits. This is a fictional, point-in-time dataset, so it can show me strong associations but it cannot prove that, say, overtime **causes** people to quit. There may be a hidden factor driving both. The numbers also describe the past, not a forecast.

The natural next steps, and the direction I'd take this if it were a live brief:

- Build a classification model (logistic regression or a gradient-boosted tree) to estimate each employee's attrition risk and confirm which factors carry weight once the others are held constant.
- Bring in time-based data so the analysis can move from “who left” to “who is about to,” which is where it becomes a tool rather than a post-mortem.
- Validate the overtime and Sales findings against real exit-interview notes, to check whether the data story matches what people actually say on the way out.

8. What's in the repository

Everything needed to follow or reproduce this project sits in the repo:

File	What it is
HR_Attrition_Analysis_EDA.ipynb	The Python notebook: cleaning, correlations and profiling
IBM_HR_Attrition_SQL.sql	Nine business questions answered in SQL
HR_Attrition_analysis_Dashboard.pbix	The Power BI dashboard file
HR_Dashboard.pdf	A static export of the dashboard
IBM_HR_Attrition_Report.docx	This report
IBM_HR_Attrition_Presentation.pptx	The slide deck version
README.md	Project overview and how to run it

Thanks for reading. If you want to discuss the approach or the findings, the contact links are in the repository's README.